

S Lab Sync — Setup Guide

Everything you need to install on a fresh computer before running the app

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Before You Start: The Checklist

S Lab Sync is a Python desktop program that stores its data in a MySQL database. Neither Python nor MySQL comes with the app, so on a brand-new computer three things need to be in place before it will run:

#	Requirement	Why it's needed
1	Python 3.10 or newer, with Tkinter	The app is a Python program. Tkinter is the toolkit that draws its windows; it is built into the Windows and macOS installers, but a separate package on Linux.
2	MySQL Server	All student, teacher, equipment, chemical and experiment data is stored in MySQL, so a MySQL server has to be running somewhere the app can reach.
3	The mysql-connector-python package	Python cannot talk to MySQL on its own. This one package is the bridge, and a single command installs it.

Each of these is a one-time setup per computer. After that, starting the app is a single command or double-click.

STEP 1 Install Python (3.10 or newer) with Tkinter

First check whether Python is already installed. Open a terminal or command prompt and run:

```
python --version (Windows: py --version)
python3 --version (macOS / Linux)
```

If it prints version 3.10 or higher, skip ahead to the Tkinter check below. Otherwise install it:

Windows

- Download the installer from **https://www.python.org/downloads/**.
- Run it and **tick "Add python.exe to PATH"** on the first screen, then choose Install Now. Tkinter (listed as "tcl/tk and IDLE") is included by default.

macOS

- Easiest: download the installer from **https://www.python.org/downloads/**. It includes Tkinter.
- With Homebrew instead: `brew install python python-tk`.

Linux (Debian / Ubuntu)

```
sudo apt update
sudo apt install python3 python3-pip python3-venv python3-tk
```

On Fedora, use `sudo dnf install python3 python3-tkinter` instead.

Check that Tkinter works

```
python -m tkinter (macOS / Linux: python3 -m tkinter)
```

A small test window should appear. If you see "No module named tkinter", Tkinter is missing — install the Tk package for your system as shown above.

STEP 2 Install MySQL Server

The app needs a MySQL server it can connect to. This is normally installed on the same computer (`localhost`), but it can also be a MySQL server elsewhere on your network.

Windows

- Download the MySQL Installer from **https://dev.mysql.com/downloads/installer/**.
- Choose **"Server only"** (or "Developer Default") as the setup type.
- When asked, create a **root password** and remember it — you will type it into the app.
- Let the installer run MySQL as a Windows service so it starts automatically.

macOS

- With Homebrew: `brew install mysql`, then `brew services start mysql`.
- Or use the official .dmg installer from the same MySQL downloads site.
- Set a root password with `mysql_secure_installation` if you were not asked for one.

Linux (Debian / Ubuntu)

```
sudo apt install mysql-server
sudo systemctl enable --now mysql
sudo mysql_secure_installation
```

Whichever system you use, note the **username** and **password** you set. The account must be allowed to create databases and tables (the default `root` account works).

STEP 3 Get the app and install its one dependency

Download the project from GitHub (**Code** then **Download ZIP**, and unzip it) or clone it with git. Open a terminal inside the project folder, then install the MySQL driver:

Windows

```
py -m pip install -r requirements.txt
```

macOS / Linux (recommended: a virtual environment)

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

> Newer Linux and macOS (Homebrew) Pythons refuse a plain "pip install" with an "externally-managed-environment" error. That is why a virtual environment (the three lines above) is recommended. Remember to run "source .venv/bin/activate" again in each new terminal before starting the app.

STEP 4 Run S Lab Sync for the first time

- **Windows:** double-click `run_slab_sync.bat`, or run `py main.py`.
- **macOS / Linux:** run `python3 main.py` (with the virtual environment active).

What to expect on first launch:

- A **"Connect to MySQL"** window appears first. Enter the host (`localhost`), port (`3306`) and the MySQL username and password from Step 2, then click **Test & Connect**. Leave *Remember connection* ticked to skip this next time.
- The app creates the `lab_management` database and all tables automatically.
- The login window appears. Sign in with username `Master` and password `7777`.
- **Change the master login straight away** using *Change Password* in the sidebar.

Troubleshooting Common First-Run Issues

Symptom	Likely cause and fix
"No module named tkinter"	Tkinter is not installed. Install `python3-tk` (Linux), `brew install python-tk` (Homebrew) or reinstall Python from python.org.
"No module named mysql"	The driver is not installed for the Python you are running. Run the pip command from Step 3 again, in the same virtual environment you start the app from.
"python is not recognized" (Windows)	Python is not on PATH. Reinstall it with "Add python.exe to PATH" ticked, or start the app with `py main.py`.
pip says the environment is "externally managed"	Use a virtual environment as shown in Step 3 instead of installing into the system Python.
"Access denied for user"	The MySQL username or password is wrong. Click *MySQL Settings* in the sidebar to enter the correct details.
"Can't connect to MySQL server" / connection refused	MySQL is not running. Start the MySQL service (Windows Services), `brew services start mysql` (macOS) or `sudo systemctl start mysql` (Linux), and check the host and port.
Forgot the master login	Delete the file `credentials.json` inside the `.labmanager` folder in your user profile. The login returns to Master / 7777.
Want to forget the saved MySQL connection	Delete `python_db.json` from the same `.labmanager` folder; the setup window will appear on the next launch.

Quick Reference

Item	Where to get it / command
Python 3.10+	https://www.python.org/downloads/
MySQL Server	https://dev.mysql.com/downloads/installer/ (or your system's package manager)
MySQL driver	pip install -r requirements.txt (installs mysql-connector-python)
Start the app	py main.py (Windows) | python3 main.py (macOS / Linux)
Default app login	Username: Master | Password: 7777 (change it after first login)
Settings folder	~/.labmanager/ (credentials.json and python_db.json)

Once Python, MySQL and the driver are in place, this setup only needs to be done once per computer — every launch after that is just starting the app.